

Problem Set 3

Upper and Lower Sums/Convergence

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Problem 1

(f)

$$M_5 = 20 + 30 + 32 + 26 + 12 = 120mC$$

$$M_5 = L_5 = 120mC$$

No, the agreement is an accident. Q does lie inside the bracket.

$$E_L = \frac{5}{3}mC$$

$$E_M = \frac{5}{3}mC$$

$$E_R = \frac{25}{3}mC$$

Problem 3

(a)

$$L_3 = R_3 = 6 + 6 + 4 = 16L$$

The agreement does not indicate that the result is accurate.

(b)

$$F' = -2t + 3$$

$$\max : (1.5, 6.25)$$

$$t_1 \implies \max : 6, \min : 4$$

$$t_2 \implies \max : 6.25, \min : 6$$

$$t_3 \implies \max : 6, \min : 4$$

$$\overline{S}_3 = 6 + 6.25 + 6 = 18.25L$$

$$\underline{S}_3 = 4 + 6 + 4 = 14L$$

(c)

$$\overline{S}_3, \underline{S}_3 \implies [14, 18.25], w = 4.25L$$

$$R_3, L_3 \implies \{16\}, w = 0L$$

The second approach assumes that the function is monotonically increasing over the interval.

(d)

$$16.5 \in [14, 18.25]$$

$$16.5 \notin \{16\}$$

(e) The second rectangle, as $F(t)$ changes monotonicity in the interval.

(f) For functions that are monotonically increasing or decreasing. Otherwise, the lower and upper Riemann sums must be considered.

Problem 4

(a)

$$\overline{S}_n - \underline{S}_n = R_n - L_n = \sum_{k=1}^n f(x_k) \Delta x - \sum_{k=1}^n f(x_{k-1}) \Delta x = \sum_{k=1}^n (f(x_k) - f(x_{k-1})) \Delta x$$

(b)

$$\overline{S}_4 - \underline{S}_4 = \Delta x \sum_{k=1}^4 (f(x_k) - f(x_{k-1})) = \Delta x (f(x_4) - f(x_0))$$

$$\overline{S}_n - \underline{S}_n = \sum_{k=1}^n (f(x_k) - f(x_{k-1})) \Delta x = (f(b) - f(a)) \Delta x = \frac{(b-a)(f(b) - f(a))}{n}$$

(c)

$$\lim_{n \rightarrow \infty} \overline{S}_n - \underline{S}_n = \lim_{n \rightarrow \infty} \frac{(b-a)(f(b) - f(a))}{n} = 0$$

(d)

$$\overline{S}_n - \underline{S}_n = \frac{(b-a)(f(b) - f(a))}{n}$$

$$\overline{S}_4 - \underline{S}_4 = \frac{(b-a)(f(b) - f(a))}{4}$$

(e)

$$\overline{S}_n - \underline{S}_n = \frac{(b-a)(f(b) - f(a))}{n} < 0.001$$

$$n > 1000(b-a)(f(b) - f(a))$$

$$\frac{\overline{S}_n - \underline{S}_n}{2} = \frac{(b-a)(f(b) - f(a))}{n} < 0.001$$

$$n > 500(b-a)(f(b) - f(a))$$

The existence of the identity reduces the amount of arithmetic needed to a minimal degree.

(f) Identity for f decreasing:

$$\overline{S}_n - \underline{S}_n = \frac{(b-a)(f(a) - f(b))}{n}$$

Problem 3's F satisfies neither, as the function is not monotonic over the whole interval. The formulas and a proper amount of arithmetic are then needed to bound the value of the function.

Problem 5

$$D(x) = \begin{cases} 1, & x \text{ rational} \\ 0, & x \text{ irrational} \end{cases}$$

(a)

$$\begin{aligned} u \in \mathbb{Q} &\implies \lim_{x \rightarrow 0+} u \in \mathbb{I} \\ u \in \mathbb{I} &\implies \lim_{x \rightarrow 0+} u \in \mathbb{Q} \\ x \in [u, v] &\implies D(x) \in \{0, 1\} \\ \sup D &= 1, \inf D = 0 \\ \overline{S}_n &= \sum_{k=1}^n 1 \cdot \frac{1}{n} = 1 \\ \underline{S}_n &= \sum_{k=1}^n 0 \cdot \frac{1}{n} = 0 \end{aligned}$$

(b)

$$\begin{aligned} S_{\mathbb{Q}} &= \sum_{k=1}^n D\left(\frac{k}{n}\right) = \sum_{k=1}^n \frac{1}{n} \cdot 1 = 1 \\ S_{\mathbb{I}} &= \sum_{k=1}^n D\left(\frac{k-1}{n} + \frac{1}{\sqrt{2}n}\right) = \sum_{k=1}^n \frac{1}{n} \cdot 0 = 0 \end{aligned}$$

The result of integrating $D(x)$ over $[0, 1]$ solely depends on the set of points chosen, rendering no single value to be called as the result.

(c)

$$\overline{S}_n - \underline{S}_n = 1 - 0 = 1$$

The difference between the sums remains as a constant no matter the size of n .

$$\underline{S}_n < S_n < \overline{S}_n$$

The range remains large and fails to be helpful towards more precise measurements.

(d)

$$\begin{aligned}
 T_n &= \frac{1}{2}(L_n + R_n) = \frac{1}{2}(\overline{S}_n + \underline{S}_n) \\
 \underline{S}_n &< A < \overline{S}_n \\
 |A - T_n| &\leq \frac{1}{2}(\overline{S}_n - \underline{S}_n) = \frac{(b-a)(f(b) - f(a))}{2n}
 \end{aligned}$$

(e)

$$\begin{aligned}
 \overline{S}_n - \underline{S}_n &= (f(b) - f(a)) \cdot \frac{(b-a)}{n} \leq \frac{V(b-a)}{n} = V\Delta x \\
 f(x) &= \sin x \implies V = 1 + 1 = 2 \\
 \overline{S}_n - \underline{S}_n &\leq \frac{2\pi}{n} \\
 \frac{2\pi}{n} &\leq \frac{1}{100} \implies n \geq 200\pi \\
 \boxed{n} &= 629
 \end{aligned}$$